

1. Assessment Style

- **Homework** – Four questions given every three weeks in advance of lecture material. Homework will be hosted on our tutorial's [Postal Owl](#) page. submissions will only be accepted through OWL as either Markdown with KaTeX math or PDF.

Submissions may be repeated even after an initial grade is awarded, but points will not be awarded to solutions submitted after the due date unless explicit permission is given *well in advance*.

- **Oral Presentation** – During the tutorial session following the due date for a homework, students will be asked to group up, formulate a collaborative solution and present it to the rest of the class. An exam-style grade will be given: 1 *excellent*, 2 *very good*, 3 *pass* and 4 *fail*.

Given our presentation structure and strict due dates, I strive to respond to homework submissions as quickly as possible. **Owl will be checked at 10:00 am every Monday, Tuesday, Thursday and Friday**; submissions made by that time will be graded and responded to within the day. **Please use this availability to ask questions early and often.** Contact Jamie if you do not yet have access to our Owl page.

2. Credit Requirement

- Earn at least 50% of the available homework points.
- Present one problem from each homework.
- Score a 4 (failure) on no more than one of said presentations.

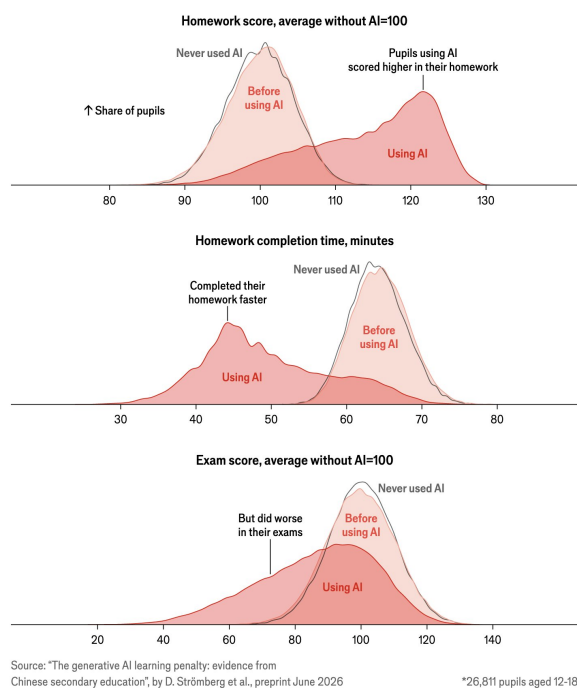
Any student who feels that they may need accommodation because of a disability (learning disability, attention deficit disorder, social anxiety disorder, etc.), please contact me as soon as possible.

3. Collaboration is Encouraged

On any assignment, you are encouraged to work with other students and to freely ask questions of the instructor online or in person. If your goal is to learn or to do well on the exam, I think it is better for you to talk with real people—other students or a human teacher—rather than to any sort of artificial intelligence.

4. Artificial Intelligence Policy

Recent research says: if you are outsourcing your homework to an AI, then you are learning less, you aren't improving your endurance, and you will struggle on the exam. I cannot monitor AI usage, but I am trying very hard to make himself available to you; please take advantage of that!



More than half of AI students spend less time than even the fastest non-AI students. This group receives very high homework scores [and] extremely low exam scores.

After six months of use, homework scores rise by 18 percent of the baseline mean while completion time per assignment falls from 64 to 45 minutes. However, students' scores in monthly school exams fall by 20 percent of the baseline mean.

Our finding that exam scores deteriorate gradually over time suggests that homework outsourcing may undermine not only the retention of current material but also the absorption of future material [2]. ... It is also consistent with recent evidence on "cognitive endurance," the capacity to sustain effortful attention in improving cognitive ability [1].

Stromberg, Lei, and Wu [3]

Algorithms and Data Structures II

Tutorial Syllabus (continued)

5. Homework Suggestions Clarity is important and will often require introducing some notation of your own. Assign precise names (A or B , x or y) to objects rather than calling them by pronouns ("this set" or "that vector"). You are strongly encouraged to learn the basics of writing your math in as it is the standard typesetting language in every mathematical field. OWL supports both Markdown and KaTeX inputs; documentation on both are linked above the editing box of the homework assignments!

<https://wordpress.com/support/markdown-quick-reference/>

<https://katex.org/docs/supported.html>

Genuine collaboration is one of the most effective ways to learn anything. I strongly encourage collaborating with peers when solving homework tasks, but genuine collaboration requires that everyone is honest and contributes to the conversation. Here are some general guidelines:

- Try to solve the task on your own before collaborating. You cannot really contribute if you haven't tried it.
- Give other students space to talk. They will have insights that you did not see.
- *Always submit your own solutions!* Duplicate submissions will be reported and treated as plagiarism.

6. Course Guidelines

- Attendance is not required and will not contribute to your grade beyond the need to give presentations in person.
- Plagiarism is strictly forbidden and will be punished with course failure and possibly with university discipline. In this instance, plagiarism includes duplicate submissions between students and AI generated submissions.
- Be respectful of your fellow students. Bullying, jeering or other disrespectful behavior will see you removed from the classroom and potentially earning a failing grade.

Week		Topic		Presentations
1	05/10	String Search	KMP	HW1: String Search
2	12/10	String Search	AC	
3	19/10	String Search	RK	
4	26/10	Network Flows	FF	HW2: Network Flows
5	02/11	Network Flows	Dinic	
6	09/11	Network Flows	PR1	
7	16/11	Fast and Parallel	FFT	HW3: Fast and Parallel
8	23/11	Fast and Parallel	Addit	
9	30/11	Fast and Parallel	PSort	
10	07/12	Geo. Algorithms	CHull	HW4: Geo and Hardness
11	14/12	Geo. Algorithms	PlSwp	
12	04/01	Hardness		

*Tuesday Tutorial Canceled

*End of Semester: 08/01

*Start of Exams: 11/01

Table 1: Tentative Calendar

References

- [1] Christina Brown et al. "Cognitive endurance as human capital". In: *The Quarterly Journal of Economics* 140.2 (2025), pp. 943–1002. DOI: [10.1093/qje/qjae043](https://doi.org/10.1093/qje/qjae043).
- [2] James J Heckman and Chase O Corbin. "Capabilities and skills". In: *Journal of human development and capabilities* 17.3 (2016), pp. 342–359. DOI: [Capabilitiesandskills](https://doi.org/10.1016/j.jhealecon.2016.03.001).
- [3] David Stromberg, Victor Lei, and Yanhui Wu. "The Generative AI Learning Penalty: Evidence from Chinese Secondary Education". In: *SIAM Journal on Computing* (2026). DOI: [10.2139/ssrn.6868618](https://doi.org/10.2139/ssrn.6868618).